



Fig. 2: Screenshot of an example of hex listing file

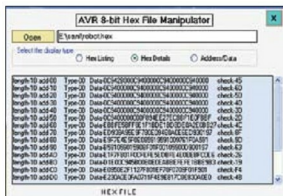


Fig. 3: Screenshot of the hex details file for the example

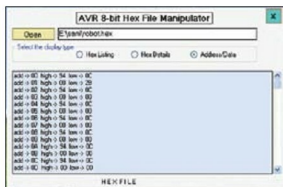


Fig. 4: Screenshot of address and data display for the example

When 'Hex Listing' or 'Hex Details' radio button is clicked, public method `readFile(string filename, bool List_details)` returns a string to the caller that displays hex listing or hex details depending on the argument passed. The returned string is a simple text file. When the `List_details` argument passed by the caller is true, the returned string contains hex details. When the `List_details` argument passed is false, the returned string contains hex listing.

Public method `parseRecord(string hexLine)` returns details of hex record format for the `hexLine` string argument passed to it. Here, only a single

line of the hex file is passed as an argument to be decoded and responded to errors, if any, using exceptions. The `readFile` method is used for manipulating each data byte in the hex file.

Public method `readFile(string filename)` is overloaded when 'Address/Data' radio button is pressed. This method returns a string containing the data to be represented as address and data for the hex file. All the three methods discussed here are defined in the `FileParser.cs` file, which is compiled to make the `HexParser.dll` file. Any program written in a .NET-supported language can use the `HexParser.dll` file.

Using this software, up to 256 kB of the compiled AVR source code can be manipulated. This capability can be augmented by changing some values in the `.dll` file. The software can also be easily modified for hex files of the 8051 MCU within the `.dll` file itself.

Designing the application

Installation of Visual Studio 2008. Visual Studio is a complete set of development tools for building ASP.NET Web applications, XML Web services, desktop applications and mobile applications. Visual Basic, Visual C# and Visual C++ all use the same integrated development environment (IDE), which enables tool sharing and eases the creation of mixed-language solutions. In addition, these languages use the functionality of the .NET Framework, which provides access to key technologies that simplify the development of ASP Web applications and XML Web services.

The essential components—.NET, Windows Installer 3.1 and others—are installed automatically. Select the customised style of installation when asked. Uncheck all the mobile-sup-

ported Smart Device development tool installations and also the SQL server 2005 contents and proceed for installation.

Launching the application. Launch the application by clicking Microsoft Visual Studio 2008 in 'Start' menu on your desktop. Create a new project. A blank form appears on the screen. It is basically a design space having close, maximize and minimize buttons, title bar, etc. Control tools required for designing the form are TextBox, OpenFileDialog button, Buttons and Radiobuttons. These can be placed on the form by dragging each control from Toolbox.

Coding and compilation. In 'Solution Explorer,' click 'form1.cs' and select 'View Code.' This opens the coding box, where you can write the code in Visual C# language and save the file. In 'Build' menu, click 'Build Solution' to compile the program code. Alternatively, you can click 'Start Without Debugging' or 'Start Debugging' from 'Debug' menu to compile and run the program.

Generating the executable file. If there is no error in the above steps, the executable file is created automatically. In 'Solution Configuration' drop-down box on the task bar, select 'Release' option to make the generated `.dll` file work properly.

Running the application

The user interface is very simple to use. Click 'Hex File Parser' application under 'Release' folder to run the application. To load the hex file, press 'Open' button in the top left-hand corner. A dialogue box opens. Select the desired hex file. To examine the hex listing file, click 'Hex Listing' radio button. Fig. 2 shows the screenshot of an example of hex listing file.

To examine the hex file details, select 'Hex Details' radio button. Fig. 3 shows the screenshot of hex details along with checkmarks. You can manually check these to confirm whether the program outputs the correct checkmarks.

To view the address and data of